

System Requirements Specification, Architecture, Wireframes & Process Flowcharts

Architectural Heritage: Harmonized design patterns from Lab 2, Lab 3, Lab 4, and Lab 5 • **Persistence:** SQLite3 ACID Engine

1. System Requirements Specification Matrix

In accordance with software engineering principles, the system is designed around balanced **Functional Requirements (FR)** and **Non-Functional Requirements (NFR)**:

Requirement Dimension	Functional Requirements (FR)	Non-Functional Requirements (NFR)
Core Definition	Define what the system should do (features & system functionality).	Define how the system should perform (quality attributes & constraints).
Operational Focus	Focus on system behavior, user interaction, and transactional operations.	Focus on performance, security, data integrity, and code maintainability.
Scope of Actions	Describes specific actions like route querying, seat booking, ticket issuance, and ticket cancellations.	Describes constraints like response times (<100ms), input sanitization, and OOP extensibility.
Visibility	Directly visible to commuters, ticket agents, and dispatchers.	Indirectly visible, governing underlying robustness, security, and architectural reliability.
Validation Metric	Easier to measure (output-based validation e.g. ticket record generated, seat reserved).	Harder to measure, verified via automated test benchmarks and strict database constraint tests.
Design Driver	Drives the core user workflows and interactive GUI features.	Influences database schema indexing (ACID SQLite), factory patterns, and class encapsulation.
Documentation Format	Documented using use cases, user stories, and operational flowcharts.	Documented using technical schemas, data dictionaries, and performance criteria.

1.1 The 2 Functional Requirements (FR)

- **FR-1: Route Schedule Browsing & Interactive Seat Selection** — The system shall allow commuters and dispatchers to browse intercity bus schedules (origin, destination, departure time, and base fare) and dynamically select an available seat through an interactive 2x2 bus cabin layout with an aisle.
- **FR-2: Booking Processing, Ticket Issuance & Lifecycle Cancellation** — The system shall process confirmed seat reservations into persistent SQLite records, generate tamper-resistant verification codes, and support ticket cancellation with immediate seat inventory release and audit logging.

1.2 The 2 Non-Functional Requirements (NFR)

- **NFR-1: Security, Validation & Tamper Resistance** — All passenger inputs undergo strict sanitization. Contact numbers must conform to standard mobile formats (09XXXXXXXXX or +639XXXXXXXXX). Concession fare discounts require mandatory verification of official institutional/government ID credentials (minimum 4 characters). Every ticket is issued with an 8-character cryptographically salted SHA-256 reference code (BTK-[HEX]) preventing counterfeiting.
- **NFR-2: Maintainability, Clean Architecture & Double-Booking Prevention** — The codebase strictly enforces Object-Oriented encapsulation (private `_` attributes, `@property` getters, and `@setter` validation raising `TypeError` or `ValueError`) conforming to PEP 8 standards. At the database layer, an ACID-compliant SQLite schema with a partial unique index (ON tickets (route_id, travel_date, seat_number) WHERE status = 'Confirmed') mathematically guarantees zero double-booking.

2. Unique System Use Case

Interactive Visual Seat Layout Matrix with Senior/PWD Priority Allocation & Concession Discount Engine (UC-SMARTBUS-01)

Actor: Commuter / Ticketing Counter Agent | **Precondition:** Routes & Fleet Buses initialized in SQLite.

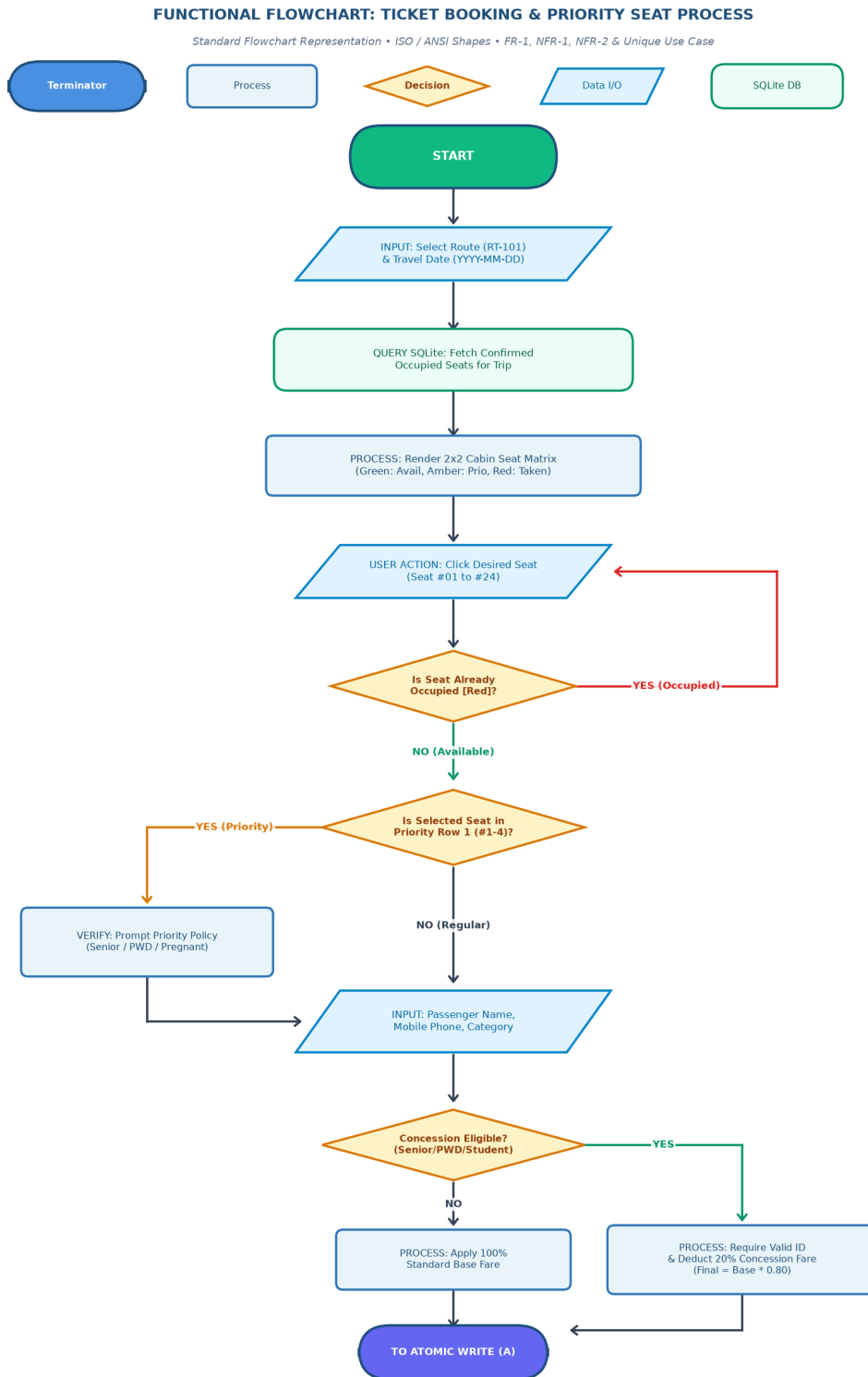
Workflow Narrative:

1. Dispatcher selects *RT-101: Manila (Cubao) → Baguio City* and travel date.
2. System draws the **Visual Cabin Seat Map**: Row 1 (Seats 1-4) is color-coded in **Amber** as **Priority Seating** (Senior Citizens, PWDs, Pregnant), while Rows 2-6 are standard **Green**.
3. Selecting an available seat highlights it in **Blue** ([✓]). Non-priority commuters choosing Row 1 trigger a confirmation alert.
4. Selecting **Senior Citizen**, **PWD**, or **Student** dynamically unlocks the Concession ID input and applies a statutory **20% discount** (Base PHP 580.00 → Net PHP 464.00).
5. Booking is committed to SQLite, seat status transitions to **Occupied [X]**, and an official printable boarding pass is issued.

3. Operational Process Flowcharts (Standard Flowchart Representation)

The following diagrams depict the operational workflows utilizing standard generic flowchart shapes: **Terminator capsules** (Start/End), **Input/Output parallelograms**, **Process rectangles**, **Decision diamonds** (Branching logic), and **Database cylinders** (SQLite queries):

3.1 Standard Flowchart: Ticket Booking & Concession Allocation (FR-1 & Unique UC)



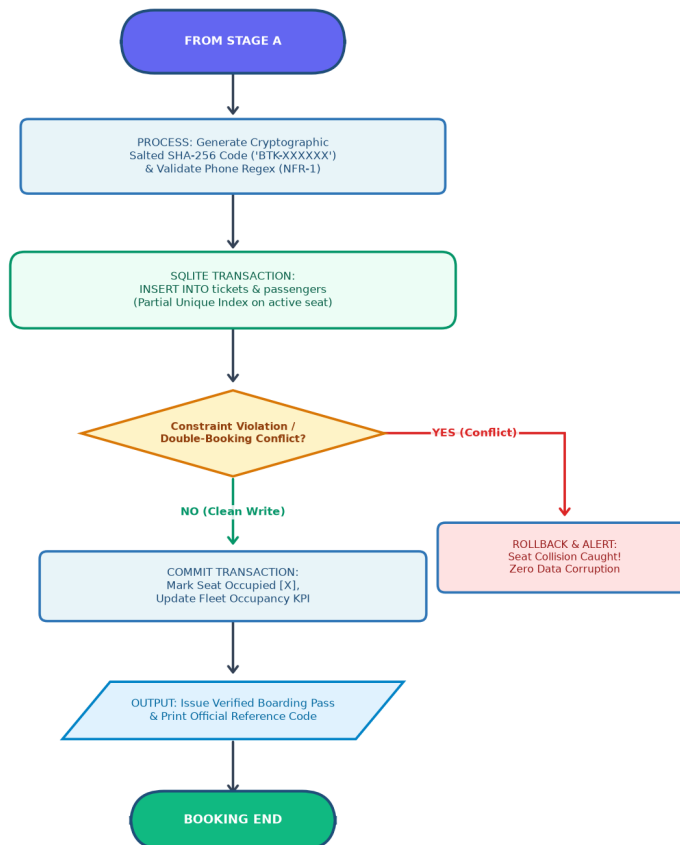
3.2 Standard Flowchart: Atomic SQLite Commit & Ticket Cancellation (FR-2 & NFR-2)

Illustrates the atomic write transaction, double-booking rollback guard, and ticket cancellation lifecycle with instant seat restoration:

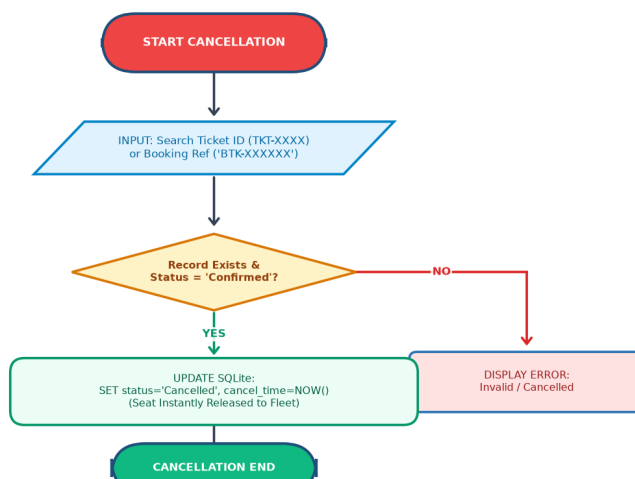
FUNCTIONAL FLOWCHART: ATOMIC COMMIT & TICKET CANCELLATION

Double-Booking Prevention (NFR-2) & Ticket Lifecycle Seat Restoration (FR-2)

[SECTION A: ATOMIC SQLITE COMMIT & VERIFICATION]

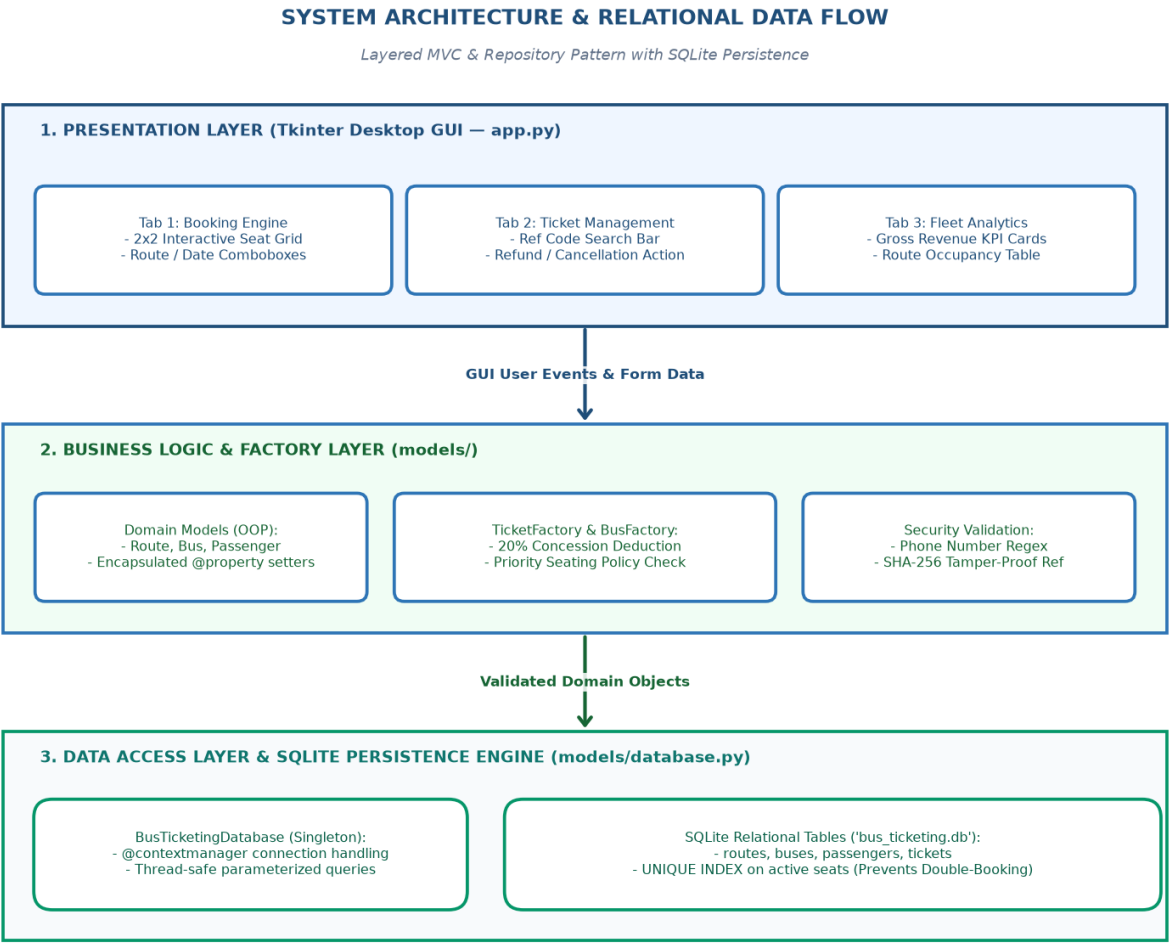


[SECTION B: TICKET CANCELLATION & SEAT RESTORATION (FR-2)]



3.3 High-Level Layered Architecture & Relational Data Flow

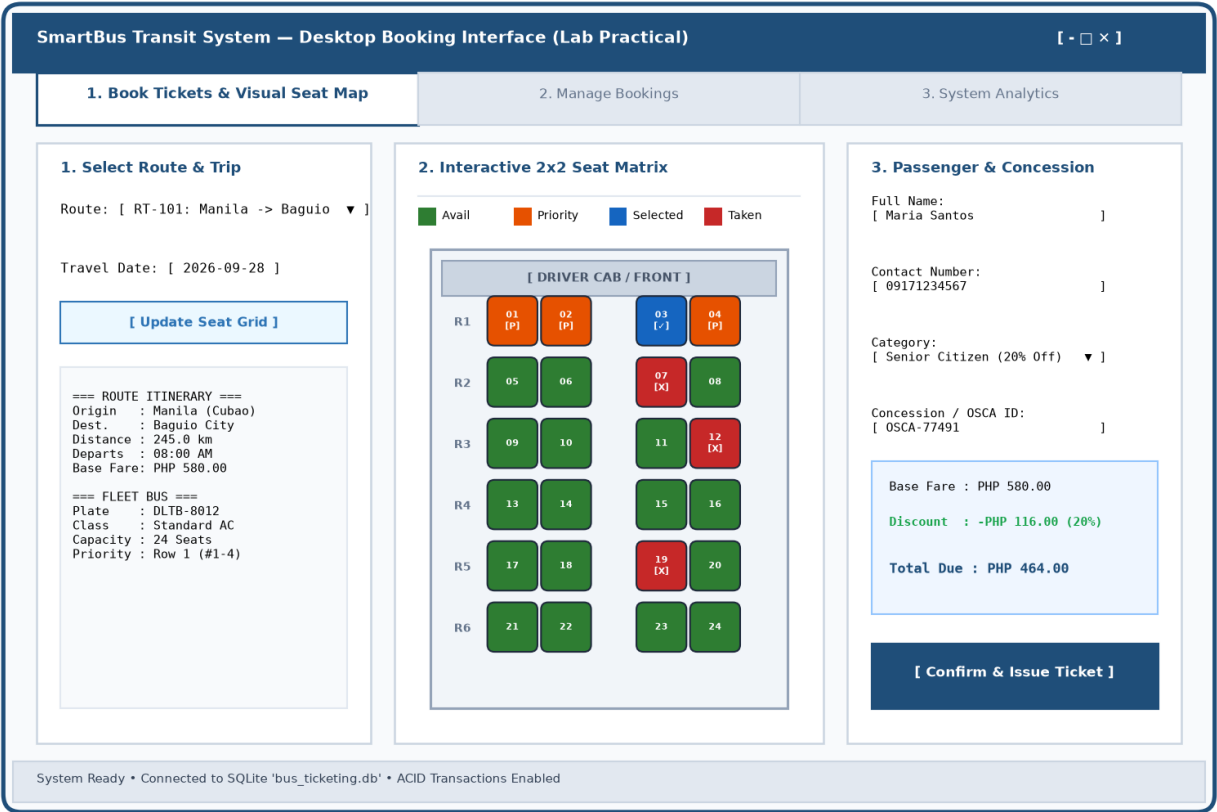
Depicts the structural separation across the Presentation Layer (Tkinter GUI), Business Logic / Factory Layer, Data Access Layer (BusTicketingDatabase Singleton), and SQLite Persistence Engine:



4. UI Wireframe Schematics & Layout Blueprints

Clean component wireframe schematics illustrating interface layout, buttons, dropdowns, tables, and the 2x2 seat map across all three tabs:

4.1 Tab 1 Wireframe: Book Tickets & Visual Seat Map



4.2 Tab 2 Wireframe: Manage Bookings & Audit Dossier

SmartBus Transit System — Ticket Lifecycle Management & Verification

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1. Book Tickets & Seat Map

2. Manage Bookings & Cancellations

3. System Analytics

Search: [TKT-1001 or BTK-XXXXXX] [Find Ticket] [Reset View] Filter: [All Statuses ▼]

Active Ticket Dossier & SHA-256 Tamper-Proof Audit (NFR-1)

Ticket ID : TKT-1001

Security Ref : BTK-8F3A29 (SHA-256)

Passenger Name : Maria Santos

Classification : Senior Citizen

Status : [CONFIRMED]

Route : Manila -> Baguio

Travel Date : 2026-09-28 08:00 AM

Concession ID: OSCA-77491

Assigned Bus: DLTB-8012 (Standard AC)

Assigned Seat: Seat #01 [PRIORITY ROW]

Fare Paid : PHP 464.00 (20% Off)

Contact No. : 09171234567

[Cancel & Refund]

[View Boarding Pass]

Master Ticket Ledger & Audit Records

Ticket ID	Reference (NFR-1)	Passenger	Category	Route	Date	Seat	Fare	Status
TKT-1001	BTK-8F3A29	Maria Santos	Senior	Manila->Baguio	2026-09-28	#01 [P]	PHP 464.00	CONFIRMED
TKT-1002	BTK-4C91E2	Juan Dela Cruz	Regular	Manila->Baguio	2026-09-28	#07	PHP 580.00	CONFIRMED
TKT-1003	BTK-1D77A0	Grace Ramos	PwD	Manila->Batangas	2026-09-28	#02 [P]	PHP 184.00	CONFIRMED
TKT-1004	BTK-77BC11	Mark Bautista	Student	Manila->La Union	2026-09-28	#05	PHP 496.00	CANCELLED

4.3 Tab 3 Wireframe: System Overview & Fleet Analytics

SmartBus Transit System — Operational Analytics & Fleet Performance

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1. Book Tickets & Seat Map

2. Manage Bookings

3. System Overview & Analytics

Total Bookings

4

Active Confirmed

3

Cancelled Tickets

1

Gross Revenue (PHP)

1,228.00

Concession Subsidies

272.00

Fleet Route Popularity & Occupancy Rates

Route ID	Origin -> Destination	Capacity	Confirmed Sold	Occupancy Rate (%)
RT-101	Manila (Cubao) -> Baguio City	24 seats	2 sold	8.3%
RT-102	Manila (Buendia) -> Batangas Port	24 seats	1 sold	4.2%
RT-103	Manila (Cubao) -> Naga City	24 seats	0 sold	0.0%
RT-104	Manila (Pasay) -> La Union (San Juan)	24 seats	0 sold	0.0%

SQLite Persistence & ACID Concurrency Audit Verification

- Persistence Status : ONLINE [SQLite Relational File: 'bus_ticketing.db']
- ACID Mode : Immediate Transactions Enabled with Foreign Key PRAGMA Validation
- Double-Booking Net : Partial Unique Index: idx_unique_active_seat ON tickets(route_id, travel_date, seat_number) WHERE status='Confirmed'
- NFR-1 Security : Salted SHA-256 Reference Code Generation ('BTK-[HEX]') & Mobile Regex Verification
- NFR-2 Quality : PEP 8 Encapsulation with Private Fields & Property Setters Raising Explicit Exceptions

4.4 Official SmartBus Boarding Pass Modal Dialog

SMARTBUS TRANSIT — OFFICIAL BOARDING PASS

GER NAME : MARIA SANTOS
RY : SENIOR CITIZEN

URE ROUTE: MANILA (CUBAO) -> BAGUIO CITY
DATE : 2026-09-28 | TIME: 08:00 AM
/ BUS : DLTB-8012 (Standard AC)
ED SEAT : SEAT #01 [PRIORITY]

TY CODE : BTK-8F3A29 (SHA-256 Verified)
STATUS : CONFIRMED
AID : PHP 464.00

arrive at terminal 30 minutes before scheduled depa

5. Verification Test Suite Matrix (12/12 Passing)

The system was validated using an automated unittest.TestCase suite testing all functional and non-functional requirements:

Test ID	Test Method Name	Category	Verification Criteria	Status
TC-01	test_01_singleton_database_instance	Architecture	Verifies BusTicketingDatabase enforces Singleton pattern across calls.	PASS
TC-02	test_02_sqlite_persistence_roundtrip	Persistence	Verifies SQLite CRUD operations for Routes, Buses, and Passengers.	PASS
TC-03	test_03_route_validation_boundaries	NFR-2	Verifies positive distance and non-negative base fare constraints.	PASS
TC-04	test_04_bus_seating_and_priority_validation	NFR-2	Verifies seating bounds and priority seat designations.	PASS
TC-05	test_05_passenger_contact_format_validation	NFR-1	Verifies mobile contact regex sanitization (09XXXXXXXXXX).	PASS
TC-06	test_06_mandatory_concession_id_validation_nfr1	NFR-1	Verifies mandatory concession ID check for Senior, PWD, and Student.	PASS
TC-07	test_07_senior_priority_seat_and_concession_discount	Unique UC	Verifies Row 1 priority seat booking with statutory 20% discount.	PASS
TC-08	test_08_regular_passenger_full_fare	FR-1	Verifies regular passengers receive standard undiscounted fare.	PASS
TC-09	test_09_tamper_resistant_reference_code_format_nfr1	NFR-1	Verifies cryptographically salted SHA-256 reference codes (BTK-XXXXXX).	PASS
TC-10	test_10_atomic_double_booking_prevention_nfr2	NFR-2	Verifies duplicate seat booking raises ValueError & rolls back transaction.	PASS
TC-11	test_11_ticket_cancellation_releases_seat_fr2	FR-2	Verifies cancelled ticket releases seat for immediate re-booking.	PASS
TC-12	test_12_analytics_calculation	Metrics	Verifies gross revenue, ticket counters, and subsidy calculations.	PASS

Unit Test Execution Console Output:

```
PowerShell Console - python -m unittest test_system -v

PS C:\Users\jusra\OneDrive\Documents\Practical> python -m unittest test_system -v

test_01_singleton_database_instance (test_system.TestBusTicketingSystem.test_01_singleton_database_instance)
Verify BusTicketingDatabase enforces Singleton pattern across calls. ... ok
test_02_sqlite_persistence_roundtrip (test_system.TestBusTicketingSystem.test_02_sqlite_persistence_roundtrip)
Verify entities are correctly persisted and retrieved from SQLite. ... ok
test_03_route_validation_boundaries (test_system.TestBusTicketingSystem.test_03_route_validation_boundaries)
Verify Route setter validations for positive distance and non-negative fare. ... ok
test_04_bus_seating_and_priority_validation (test_system.TestBusTicketingSystem.test_04_bus_seating_and_priority_validation)
Verify Bus seating capacity limits and priority seat definitions. ... ok
test_05_passenger_contact_format_validation (test_system.TestBusTicketingSystem.test_05_passenger_contact_format_validation)
Verify mobile phone number formatting rules (NFR-1). ... ok
test_06_mandatory_concession_id_validation_nfr1 (test_system.TestBusTicketingSystem.test_06_mandatory_concession_id_validation_nfr1)
Verify Senior Citizen, PWD, and Student must provide valid concession ID (NFR-1). ... ok
test_07_senior_priority_seat_and_concession_discount (test_system.TestBusTicketingSystem.test_07_senior_priority_seat_and_concession_discount)
Verify Unique Use Case: ... ok
test_08_regular_passenger_full_fare (test_system.TestBusTicketingSystem.test_08_regular_passenger_full_fare)
Verify regular passenger receives zero discount. ... ok
test_09_tamper_resistant_reference_code_format_nfr1 (test_system.TestBusTicketingSystem.test_09_tamper_resistant_reference_code_format_nfr1)
Verify reference codes are non-empty, uppercase, and follow 'BTK-[HEX]' format (NFR-1). ... ok
test_10_atomic_double_booking_prevention_nfr2 (test_system.TestBusTicketingSystem.test_10_atomic_double_booking_prevention_nfr2)
```